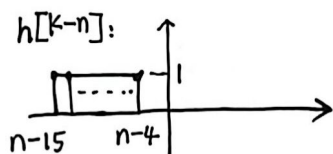
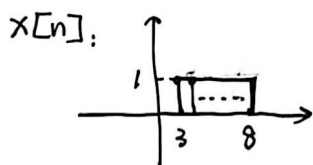


2-4.

$$y[n] = x[n] * h[n].$$



①. $3 \leq n-4 \leq 8$. 即 $7 \leq n \leq 12$.

$$y[n] = n-4-3 = n-7.$$

②. $12 < n$, $n-15 \leq 3$, 即 $12 < n \leq 18$.

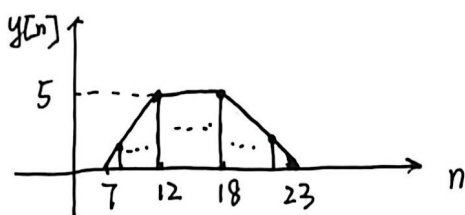
$$y[n] = 8-3 = 5.$$

③. $18 < n \leq 23$.

$$y[n] = 8 - (n-15) = 23-n.$$

④. 0-10. $y[n] = 0$.

$\therefore y[n]$:



2-11.

$$x(t) = u(t-3) - u(t-5). \quad h(t) = e^{-3t} u(t).$$

(a). $y(t) = x(t) * h(t)$.

$$\begin{aligned} e^{-3t} u(t) * u(t) &= \int_0^t e^{-3\tau} d\tau = -\frac{1}{3} e^{-3\tau} \Big|_0^t = \left(-\frac{e^{-3t}}{3} + \frac{1}{3} \right) u(t). \\ \therefore x(t) * h(t) &= -\frac{1}{3} (e^{-3(t-3)} - 1) + \frac{1}{3} (e^{-3t} - e^{-3(t-5)}) u(t). \\ &= -\frac{1}{3} (e^{-3(t-3)} - e^{-3(t-5)}) u(t). \end{aligned}$$

$\therefore t \leq 3$ 时, $y(t) = 0$.

$3 < t < 5$ 时, $y(t) = -\frac{1}{3} (e^{-3(t-3)} + 1)$.

$t \geq 5$ 时, $y(t) = -\frac{1}{3} (e^{-3(t-3)} - e^{-3(t-5)})$.

$$\therefore y(t) = -\frac{1}{3} (e^{-3(t-3)} + 1) u(t-3) + \left[-\frac{1}{3} (e^{-3(t-5)} + 1) u(t-5) \right]$$

(b). $\frac{dx}{dt} * h(t)$.

$$= [\delta(t-3) - \delta(t-5)] * e^{-3t} u(t)$$

$$= e^{-3(t-3)} u(t-3) - e^{-3(t-5)} u(t-5)$$

$$= \frac{dy(t)}{dt}.$$

(c). $g(t) = \frac{dy(t)}{dt}$.

联系方式: _____



2.17.

$$y'(t) + 4y(t) = x(t).$$

$$(a). x(t) = e^{(-1+3j)t} u(t).$$

$$\textcircled{1} \text{ 特征方程: } \lambda + 4 = 0, \lambda = -4.$$

$$y_h(t) = Ae^{-4t}$$

$$y_h(0) = A = 1.$$

$$\therefore y_h(t) = e^{-4t} u(t).$$

$$\textcircled{2} \text{ 冲激响应 } g(t) = Be^{-4t}.$$

$$g(0) = B = 1.$$

$$\therefore g(t) = e^{-4t}.$$

$$\textcircled{3} y_{zs} = \int_0^t e^{(-1+3j)(t-\tau)} e^{-4(t-\tau)} d\tau$$

$$= \int_0^t e^{(3+3j)\tau} \cdot e^{-4\tau} d\tau$$

$$= \frac{1}{(3+3j)e^{4t}} (e^{(3+3j)t} - 1) u(t).$$

$$= \frac{1-j}{6} (e^{(-1+3j)t} - e^{-4t}) u(t).$$

(b).

$$e^{-t} \cos(3t) u(t) = \operatorname{Re}(e^{(-1+3j)t} u(t)).$$

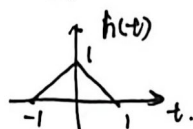
$$\therefore y(t) = \operatorname{Re}(y_{zs})$$

$$= \operatorname{Re} \left(\frac{1-j}{6} (e^{-1 \cos 3t + j e^{-1 \sin 3t} - e^{-4t}}) u(t) \right)$$

$$= \left[\frac{1}{6} (e^{-1 \cos 3t - e^{-4t}}) + \frac{1}{6} e^{-1 \sin 3t} \right] u(t)$$

$$= \left(\frac{1}{6} (e^{-1 \cos 3t + e^{-1 \sin 3t} - e^{-4t}}) \right) u(t).$$

2.23.

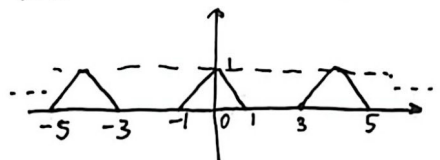


$$x(t) = \sum_{k=-\infty}^{+\infty} \delta(t - kT).$$

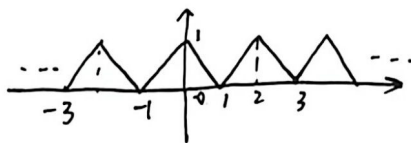
$$(a). x(t) = \sum_{k=-\infty}^{+\infty} \delta(t - 4k).$$

$$x(t) * h(t)$$

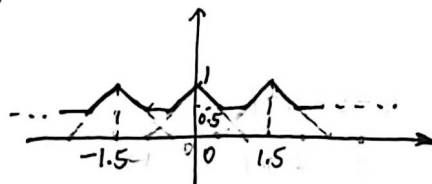
$$= \sum_{k=-\infty}^{+\infty} \delta(t - 4k) * h(t) = \sum_{k=-\infty}^{+\infty} h(t - 4k).$$



(b).



(c).



(d).

